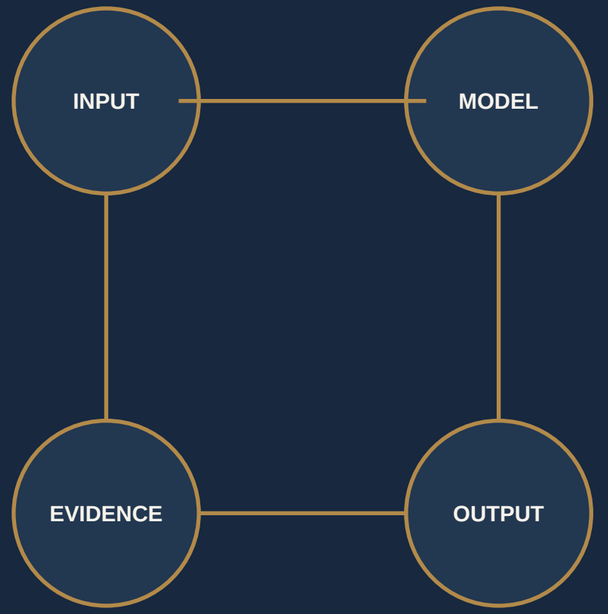
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

# Robot Manipulation

AI.405 | University lecture series



# What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define robotic arms.
- Apply key methods including grippers.
- Build or analyze a practical system involving grasping.
- Evaluate assumptions, limitations, performance and responsible use.

# A connected learning journey

Each stage prepares the tools needed for the next.

01

## FOUNDATIONS

Robotic arms

02

## METHODS

Grippers

03

## SYSTEMS

Grasping

04

## PRACTICE

Laboratory, evaluation and project

# The subject at a glance

A map of the ideas and techniques in this course

### 01 · Robotic arms
Robotic arms

### 02 · Grippers
Grippers

### 03 · Grasping
Grasping

### 04 · Force control
Force control

### 05 · Manipulation planning
Manipulation planning

### 06 · Object handling and dexterou
Object handling and dexterous manipulation

### 07 · Mandatory practical: Simulat
Mandatory practical: Simulate a robot arm performing pick-and-place

### 08 · Grasping and manipulation task
Grasping and manipulation tasks

# Core ideas, made explicit

AI.405 • Distinguish definitions, assumptions and evidence

- Robotic arms
- Grippers
- Grasping
- Force control
- Manipulation planning

**DEFINE → MODEL → TEST → INTERPRET**

# Core ideas, made explicit

AI.405 • Distinguish definitions, assumptions and evidence

- Object handling and dexterous manipulation
- Mandatory practical: Simulate a robot arm performing pick-and-place
- Grasping and manipulation tasks

**DEFINE → MODEL → TEST → INTERPRET**

# Follow the system through its stages

Use the process diagram to connect components and outcomes.

**PROBLEM**
Robotic arms

**INPUTS**
Grippers

**METHOD**
Grasping

**OUTPUT**
Force control

**EVALUATION**
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

# From concept to a real engineering decision

A short case study for discussion and worked-example narration

### SCENARIO

A real task involving robotic arms.

### CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

### DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

# Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using robotic arms.
- Implement or simulate a baseline using grippers.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

# Review · explain · apply

## KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

## PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

**Key takeaway: make the model visible, test it, and communicate its limits.**

Course-series slide design • detailed narration, demonstrations and assessments expand each module.